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A data-filtered sentiment analysis model for economic forecasting

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ABSTRACT

In behavioral economics, sentiments influence decision-making processes, with positive sentiments tending to underestimate risks and negative sentiments overestimate them. At the individual level, these sentiments shape economic behavior in ways that collectively influence broader economic dynamics. With the proliferation of textual data and advancements in natural language processing techniques, the analysis of economic sentiments has garnered growing attention, with large language models showing superior analytical capabilities. However, unlike many machine learning tasks where true labels are available through human annotation, sentiment analysis encounters challenges in obtaining true labels due to the psychological biases and inconsistencies inherent in human assessments. To address this issue, this study introduced a data filtering methodology to enhance data reliability and developed a robust sentiment analysis model tailored to the Japanese economy. The findings revealed that our model not only outperforms existing models in terms of generalization capability across diverse datasets — achieving RMSE values of 0.09–0.11 and classification accuracies of 0.83–0.88 — but also effectively captures fluctuations in other quantitative economic indicators, as evidenced by Euclidean distances of up to 1.56, which is smaller than the records 4.33 and 4.24 of the existing models. Moreover, a statistically significant correlation between qualitative and quantitative economic indicators was identified, highlighting the potential of qualitative indicators in predicting economic conditions.

1. Introduction

Behavioral economics posits that the behaviors of economic agents, such as consumers, can be interpreted by highlighting their psychological dimensions [1]. These behaviors significantly impact economic dynamics. Traditionally, economic evaluations have predominantly utilized quantitative indicators, such as Gross Domestic Product (GDP) and employment statistics, due to their measurability and objectivity. For instance, a declaration that "GDP grew by 3%" provides a clear and concise narrative that is easily understood. However, the process of generating quantitative indicators, from initial data collection to the calculation of final values, involves considerable manual labor and time. This intensive resource requirement does not align with the needs of the contemporary information society, which emphasizes the availability of real-time data. Additionally, these indicators often fail to account for the influence of market sentiment and customer behavior, both of which can significantly influence the economy [2]. As such, although quantitative indicators have historically been central, the growing availability of extensive textual data and advancements in natural language processing (NLP) technologies have enhanced the appreciation of psychological dimensions, which can now be analyzed in real time. Numerous research have successfully examined the relationship between sentiment and behavior patterns across various disciplines, including event prediction [3], health issues [4], and politics [5]. Conversely, accurately identifying human subjective feelings through sentiment analysis is undoubtedly valuable for advancing artificial intelligence technology [6].

Emotional responses, which often supersede rational thought and influence decisions based on feelings rather than logic, provide key insights for a deeper understanding of economic trends and market dynamics [7–9]. For instance, Shapiro et al. (2022) [10] identified the impulse responses of macroeconomic variables to sentiment shocks, demonstrating that positive sentiment shocks increase consumption, output, and interest rates while reducing inflation. Following a comprehensive overview, Das et al. (2024) [11] underscored the benefits and potential of utilizing market sentiment as a critical factor in stock price prediction. Sentiment analysis serves as a promising tool for governments and authorities to monitor online public opinion [12], facilitating improved risk management and response strategies, especially during emergencies or major events [13]. According to [7,14], the prices of financial assets in investment markets are continuously affected by a myriad of real-time information streams, including global economics, businesses, political, and societal news, as well as the

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sentiments with which individuals perceive this information. These factors are instantaneously reflected in the fluctuations of financial asset prices. For instance, Tetlock (2007) [15] demonstrated a correlation between mass media coverage and stock market movements; Tetlock et al. (2008) [16] revealed a relationship between the frequency of pessimistic words in articles and the accounting earnings and stock returns of individual companies. Similarly, Boudoukh et al. (2012) [17] examined that news published in newspapers impacts stock price movements. Moreover, the utilization of machine learning to make calculated predictions has proven to be highly effective in financial market by minimizing human effort and providing optimum results [18–20]. Ishijima et al. (2015) [21] performed a sentiment analysis of the Japanese economy using daily newspaper articles, developing a Sentiment Index by quantifying the frequency of words describing daily economic conditions in either positive or negative terms. Their analysis revealed shifts in the relationship between the Sentiment Index and the Nikkei 225, demonstrating a statistically significant change in the Sentiment Index across pre- and post-crisis periods, with predictive capability for stock prices up to three days in advance. Moreover, Seki et al. (2022) [22] introduced a business sentiment index that effectively reflects macroeconomic conditions, as corroborated by quantitative economic indicators. The analysis by [23] indicated that the subsequent shift in sentiment was particularly pronounced in EU countries that experienced the most severe economic impact. Moreover, the study suggests that sentiment influences the real economy in multiple ways.

On the other hand, sentiment analysis benchmark datasets are essential for the development and performance evaluation of machine learning models. Several datasets are widely recognized and utilized. For instance, the Financial PhraseBank dataset, which is tailored to the financial domain, contains approximately 4,000 phrases classified as positive, negative, or neutral [24]. While other datasets may not be specifically designed for the financial domain, they remain instrumental for economic sentiment analysis. Notable examples include the Internet Movie Database (IMDb) Movie Reviews dataset, which contains 50,000 reviews classified as positive or negative [25]; the Amazon Reviews dataset, which contains 571.54 million product reviews annotated with star ratings ranging from 1 to 5 [26]; and the Stanford Sentiment Treebank dataset, which provides sentiment labels (positive or negative) for 215,154 phrases in 11,855 sentences [27]. More open source sentiment data can be found in [28].

In Japan, the Economy Watchers Survey is one of the most esteemed financial benchmark datasets. Conducted monthly since January 2000 by a governmental agency, this survey gathers input from participants in industries particularly sensitive to economic trends. It offers insights into various economic aspects such as household dynamics, business activities, and employment conditions. The data accrued from this survey is substantial, characterized by sentiment labels related to economic conditions and textual reasoning, thereby distinguishing it globally. Consequently, it has been extensively utilized in research to analyze economic sentiment. For instance, Yamamoto et al. (2022) [29] developed a sentiment classification model to discern positive and negative economic sentiments using this dataset. Aoshima and Nakagawa (2019) [30] trained a bidirectional encoder representations from transformers (BERT, [31]) model with this data and performed sentiment analysis on Japanese Wikipedia data. Moreover, an official economic indicator, the Diffusion Index (DI), is derived from the economic sentiments of the survey participants. This index currently serves as a standard indicator of Japan's economic conditions.

However, unlike many machine learning tasks where true labels are available through human annotation, such as image recognition (e.g., classifying images into categories like "dog" or "cat") and medical diagnosis (e.g., diagnosing diseases based on symptoms and medical data), sentiment analysis is challenged by psychological biases and inconsistencies inherent in human assessments, which compromise the reliability of human annotations. Hartmann et al. (2023) [32] highlighted the importance of accurate sentiment annotation for realizing the full potential of sentiment analysis. Ensuring proper data labeling warrants sufficient attention prior to analysis, as it is among the key challenges in developing new and improved methods for textual sentiment analysis [33,34]. Inadequate annotation of text documents can lead to substantial misinterpretations [35] and diminish explanatory power [36], potentially resulting in considerable business loss [37]. Furthermore, many studies have shown that dataset noise [38], which can be introduced from human psychological biases, degrades predictive performance. Properly addressing such noise during the pre-processing stage can substantially improve model performance.

Thus, ensuring the accuracy of sentiment annotation remains a crucial concern in sentiment analysis research, a challenge that extends to large language models (LLMs), which have demonstrated impressive performance on a wide range of NLP tasks. According to [39], although LLMs in zero-shot settings still lag behind smaller language models (SLMs) trained on in-domain data, they consistently outperform SLMs when only a limited quantity of annotated data is available. Note that the design of the prompt is another critical factor leading to large performance variance of LLMs [40,41]. Zheng (2025) [42] investigated the inconsistencies and psychological biases in human economic assessments by applying behavioral economics theories, complemented by an empirical analysis of real-world data. The theories examined included *bounded rationality* (the reliance on heuristics and rules of thumb due to limitations in information-processing capabilities), *herding behavior* (the tendency to conform to the majority rather than independently analyze vast and constantly changing market information), *halo effect* (the disproportionate influence of prominent characteristics on the evaluation of an object), and *anchoring effect* (the propensity to heavily rely on the initial piece of information encountered, even if it is unrelated or irrelevant) . The experimental results indicated a correlation between declining economic indicator values and an increase in participant responses, reflecting heightened public engagement during periods of economic uncertainty. Moreover, a divergence exceeding 60% was identified between human annotations and machine predictions was identified, highlighting substantial biases that undermine the reliability of benchmark data and the sentiment models trained on them.

As noted above, previous studies in economic sentiment analysis have demonstrated that sentiment indices derived from textual data can correlate with key economic indicators and, in some cases, predict short-term market movements. However, most approaches rely directly on human-annotated benchmark datasets, which are often affected by psychological biases and inconsistencies, thereby limiting model reliability and generalization capability. Moreover, few studies have systematically integrated qualitative sentiment indicators with quantitative economic measures to enhance the timeliness and robustness of economic forecasting.

To address these gaps, this study challenges the conventional reliance on benchmark datasets by introducing a data-filtering methodology designed to enhance the reliability of sentiment annotations. Utilizing this high-quality dataset which is capable of capturing more representative and consistent patterns of economic sentiment, a robust sentiment analysis model tailored to the Japanese economy is developed, demonstrating superior performance across diverse datasets. Furthermore, qualitative and quantitative economic indicators are integrated, and their statistically significant relationship is established, offering new insights into the extent to which qualitative measures can improve the timeliness and accuracy of economic assessments.

The structure of this article is as follows: Section 2 provides a detailed description of the applied data and model framework; Section 3 outlines the evaluation experiments; Section 4 investigates the relationship between quantitative and qualitative economic indicators; and Section 5 concludes the study, offering directions for future research.

Table 1
Example responses from the Economy Watchers Survey data (translated). The table shows the economic sector, participants' location and occupation, their assessment of current/future economic conditions, the stated reason, and a translated description.

Economic sector	Region	Occupation	Economic condition	Reason for the assessment	Description
Household activity	Hokkaido	Shopping Street (Representative)	No change	Customer behavior	In the first half of this month, sales increased compared to the previous year, driven by the success of the sales campaign. However, in the latter half of the month, customer behavior indicated a more cautious approach, with individuals taking ample time to make decisions and purchasing only essential items.
Corporate performance	Hokkaido	Food Manufacturing Industry (Association Executive)	Slightly worse	Order and sales volumes	This month's sales volume has decreased compared to previous years, accompanied by intensified price competition.
Employment conditions	Hokkaido	Staffing Agency (Employee)	Slightly improved	Job openings	The demand for talent from companies remains strong, with the number of job openings at our company increasing by 50% compared to the previous year, reaching levels similar to those observed three months ago. However, while HR personnel express urgency in recruitment, many also report a decline in the quality of candidates. Companies that are unable to secure the necessary workforce are facing challenges in their efforts to expand operations.

Table 2
Number of response sentences by assessment category for the current and future economic conditions from the Economy Watchers Survey (January 2002–June 2024).

	Current condition (%)	Future condition (%)
Greatly Improved	10,462 (2.12)	9,688 (2.00)
Slightly Improved	101,884 (20.65)	102,809 (21.22)
No Change	209,859 (42.53)	224,770 (46.39)
Slightly Worse	121,603 (24.64)	106,312 (21.94)
Greatly Worse	49,652 (10.06)	40,900 (8.44)
Total	493,460 (100)	484,479 (100)

2. Data and model framework

2.1. Data

The dataset utilized in this study comprises data from the Economy Watchers Survey, spanning from January 2002 to June 2024, covering a total of 270 months. The year 2002 marks the availability of seasonally adjusted DI data, which justifies its selection as the starting point for the dataset. Moreover, since 2001, the survey has consistently engaged 2,050 participants across a broad spectrum of industries and professions that are intricately linked to economic activities. These participants are required to provide insights into household activity, corporate performance, employment conditions, and overall economic conditions.

The dataset specifically includes pairs of economic sentiment rated on a five-point scale — ranging from "greatly improved" to "greatly worse" — accompanied with the reasons provided by participants. An example of this data structure is presented in Table 1. For the calculation of the DI, each sentiment category is assigned a predefined score ("greatly improved"=1, "slightly improved"=0.75, "no change"=0.5, "slightly worse"=0.25, "greatly worse"=0). These scores are then weighted by the proportion of responses in each category. This methodological approach offers a comprehensive longitudinal perspective on trends in economic sentiment and establishes a solid foundation for analyzing sentiment and decision-marking behaviors within the Japanese economy.

Table 1 illustrates that most descriptions comprise multiple sentences reflecting varied sentiments. Thus, this study employs a sentence-by-sentence analysis to accurately capture the nuanced variations in sentiments and tones. Table 2 summarizes the data, detailing the number of response sentences categorized by assessments of current and future economic conditions. In total, the data contains 493,460 sentences describing current conditions and 484,479 sentences regarding future conditions. The most common response for both temporal categories is "no change", constituting 43% and 46% of the responses, respectively. Responses indicating minor changes ("slightly improved" or "slightly worse") collectively represent about 40%–45%, underscoring the respondents' sensitivity to subtle economic shifts. Extreme responses ("greatly improved" or "greatly worse") are less frequent, representing about 10%–12%, with "greatly improved" being the least common at approximately 2%. This dataset is characterized by its large size, multi-class structure, and imbalanced distribution.

2.2. Model framework

This section delineates the development of a model tailored to analyze sentiments pertaining to the Japanese economy. In contrast to prior studies that directly utilized the Economy Watchers Survey data, this study initiated with the extraction of reliable data as a precursor to model development. The methodology adopted is depicted in Fig. 1 and encompasses two principal phrases: data extraction and model development.

2.2.1. High-confidence data extraction

During the data extraction phase, a Japanese-specific sentence embedding model, general luke-based contrastive sentence embedding (GLuCoSE, [42]) were used to vectorize each sentence (Eq. (1)).

$$\boldsymbol{v}_i = f_{\mathrm{GLuCoSE}}(s_i) \in \mathbb{R}^d \tag{1}$$

where s_i is the ith textual description and d is the embedding dimension ($d = 768$).

Then the K-nearest neighbors (KNN) with cosine similarity was used to predict the sentiment labels of each sentence, and the label agreement criterion was defined as Eq. (2). The optimal K was selected from $C = [1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50]$ using the Elbow Method. The error rate decreased significantly as K increases up to around 10 and stabilized for both current and future economic conditions data.

$$y_i^{\mathrm{KNN}} = \arg\max_{c \in C} \sum_{j \in \mathcal{N}_k(i)} \mathbb{I}(y_j = c) \tag{2}$$

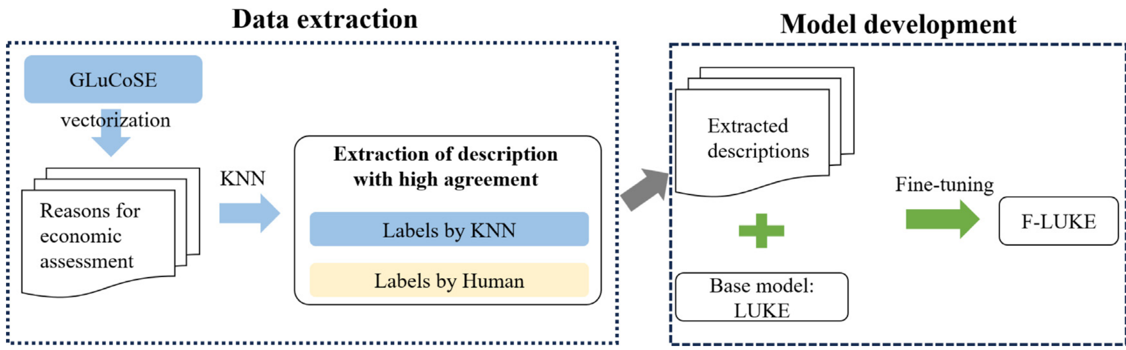


Fig. 1. Model framework. Economic assessment descriptions are vectorized using the GLuCoSE model, and high-agreement data are extracted by comparing KNN-predicted labels with human annotations. The LUKE model is the base architecture, which is fine-tuned using the extracted descriptions to develop the proposed F-LUKE model.

Sentences for which the predicted labels aligned with human-annotated labels were extracted as high-confidence data $\mathcal{H}$, which is defined as Eq. (3).

$$\mathcal{H} = \left\{ i \,\middle|\, y_i^{\mathrm{KNN}} = y_i^{\mathrm{Human}} \right\} \tag{3}$$

The proportions of high-confidence data for current and future economic conditions were 55.72% (274,961 out of 493,460) and 61.26% (296,806 out of 484,479), respectively, as shown in Table 3.

To evaluate the accuracy of the extraction process for high-confidence data, 500 response sentences were randomly sampled from each category of the five-point scale. These sentences were transformed into embeddings, and the k-means algorithm was employed to form five clusters. The silhouette score was then computed for each cluster, which evaluates the cohesion of each data point within its cluster and its separation from other clusters. A score closer to 1 suggests a strong association of the data point with its cluster and clear delineation from others.

We conducted 100 rounds of random sampling and computed silhouette scores for both the original data and the extracted high-confidence data. The student's t-test was performed to analyze the differences in silhouette scores between these two datasets. The results are presented in Table 4, and a visualization of one instance of high-confidence data using t-stochastic neighbor embedding (t-SNE, [43]) is shown Fig. 2. For both current and future economic datasets, clustering of high-confidence data exhibited significantly higher silhouette scores than those based on human annotations (current: 0.3702 *vs.* 0.3899, $p < 0.001$; future: 0.3686 *vs.* 0.3927, $p < 0.001$), indicating an enhancement in data quality. Moreover, compared to the human-annotation data, the clusters shown in the high-confidence data were visually more cohesive.

2.2.2. Model development

During the model development phase, we employed the Language Understanding with Knowledge-based Embeddings (LUKE, [44]) as the pre-trained model. Thus, the proposed model was named Financial LUKE (F-LUKE). LUKE is a language model that extends RoBERTa with an entity-aware self-attention mechanism. This feature allows the attention scores, which represent the relationships between words, to be dynamically adjusted based on specific word pairings within context. Such a design enables LUKE to capture word relationships with greater precision. Yamada et al. (2020) [44] reported that LUKE achieved state-of-the-art performance in several tasks, including Open Entity (entity recognition), TACRED (relation extraction), CoNLL-2003 (named entity recognition), ReCoRD (reading comprehension with commonsense reasoning), and SQuAD 1.1 (Stanford question answering dataset).

In this study, three sets of high-confidence data were used to fine-tune LUKE: data on current economic conditions, data on future economic conditions, and a combined data that merged current economic conditions with extreme scenarios ("greatly improved" and "greatly worse") from future economic conditions. The creation of the combined data aimed to mitigate issues related to data imbalance. Each dataset was divided into 60% for training, 20% for validation, and 20% for testing, as detailed in Table 5. The total number of samples increases from 274,961 in the current scenario to 296,806 in the future scenario, with the combined dataset containing 282,934 samples.

All samples underwent tokenization using the BERT tokenizer, with a maximum token length set to 512 tokens. The fine-tuning process spanned over 5 epochs with a learning rate of 2e-5 and a batch size of 32. The mean squared error (MSE) was employed as the evaluation metric to determine the best-performing model. Early stopping, guided by validation loss, was applied with a patience of 3 epochs. The model training was performed on a single NVIDIA GeForce RTX 4080 SUPER GPU using mixed precision training and implemented in PyTorch 2.4.0 alongside the Hugging Face Transformers library.

The model's predictions on test data were evaluated both as regression and classification tasks. For the regression task, root MSE (RMSE) was employed as the evaluation metric, using the predicted continuous sentiment scores and the discrete label (0, 0.25, 0.5, 0.75, 1). For the classification task, continuous predictions were mapped to the closest discrete label by calculating the absolute difference between each predicted value ($\hat{y}_i$) and the discrete labels. The label with the smallest difference was assigned as the predicted class label. Subsequently, precision, recall, and F1-scores were computed for each discrete label, while overall accuracy and macro-average scores were computed for comprehensive evaluation. All evaluation metric values are presented in Tables 6–8.

Tables 6–8 showed that model developed using the combined data performs robustly in both regression and classification tasks, with noticeable improvements in classification metrics. Categories such as "greatly improved" exhibited lower precision, recall and F1-scores for both current (0) and future economic conditions data (0.46, 0.32, 0.38), highlighting the challenges of accurately predicting rare classes. Nonetheless, the inclusion of extreme scenarios enhanced prediction accuracy for these rare classes by 22 (Recall: from 0.32 to 0.54) to 72 (Precision: from 0 to 0.72) percentage points. Moreover, the overall accuracy remained high at 0.83, while the macro-average metrics significantly improved, reaching 0.79, 0.75, and 0.77, indicating better-balanced predictions across all categories.

3. Evaluation

3.1. Data

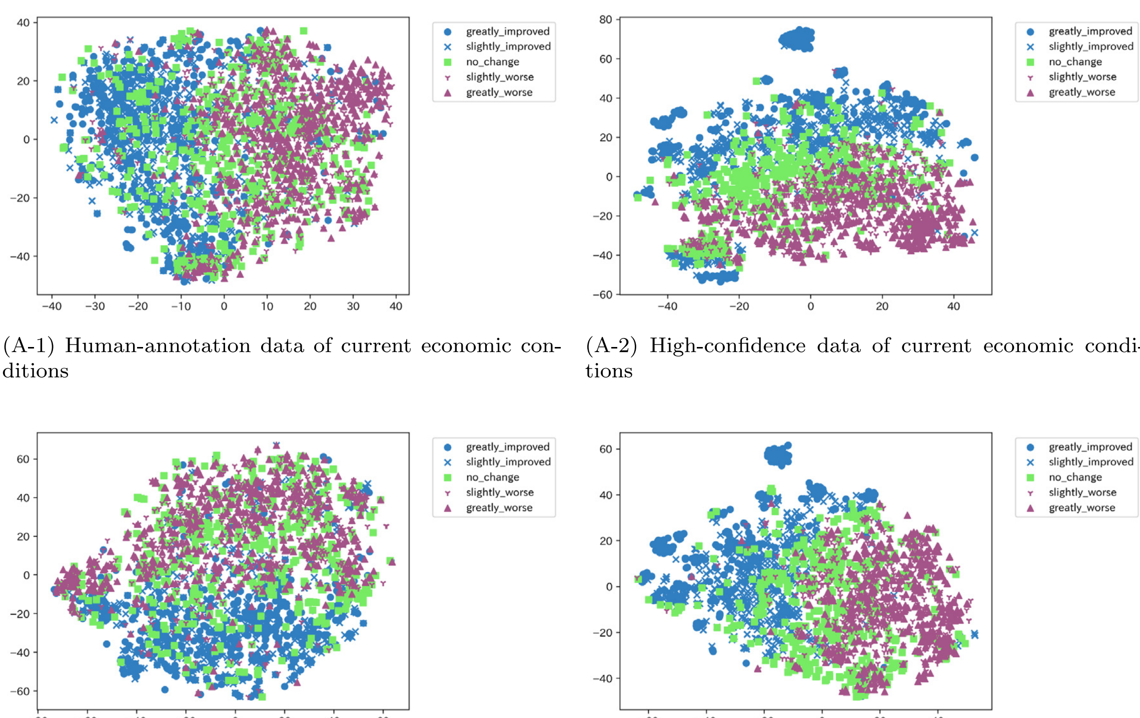
For the model evaluation experiments, two datasets reflecting the state of the Japanese economy from January 2002 to June 2024 were used: the Monthly Economy Watchers Survey data (hereafter "*watcher*") and the Monthly Economic Report (hereafter "*mer*"), both

Table 3
Distribution of responses regarding current and future economic conditions, categorized by human annotations, KNN-assigned labels, and high-confidence labels. High-confidence labels represent the subset of responses where human and KNN labels matched.

	Category	Human-annotated label (%)	KNN-assigned label (%)	High-confidence label (%)
Current economic conditions	Greatly Improved	10,462 (2.12)	1,399 (0.28)	607 (0.22)
	Slightly Improved	49,652 (10.06)	27,127 (5.50)	12,133 (4.41)
	No Change	209,859 (42.53)	225,322 (45.66)	134,042 (48.75)
	Slightly Worse	101,884 (20.65)	116,308 (23.57)	70,823 (25.76)
	Greatly Worse	121,603 (24.64)	123,304 (24.99)	57,366 (20.86)
	Total	**493,460 (100)**	**493,460 (100)**	**274,961 (100)**
Future economic conditions	Greatly Improved	9,688 (2)	1,356 (0.28)	676 (0.23)
	Slightly Improved	40,900 (8.44)	17,461 (3.6)	7,297 (2.46)
	No Change	224,770 (46.39)	250,256 (51.65)	163,655 (55.14)
	Slightly Worse	102,809 (21.22)	118,406 (24.44)	78,584 (26.48)
	Greatly Worse	106,312 (21.94)	97,000 (20.02)	46,584 (15.7)
	Total	**484,479 (100)**	**484,479 (100)**	**296,806 (100)**

Table 4
Comparison of clustering performance (Silhouette scores) between the original human-annotated data and the extracted high-confidence data for current and future economic conditions. $N = 100$ for each condition. Abbreviations: SD = Standard Deviation.

	Current economic conditions		Future economic conditions	
	Human annotation Mean (SD)	High-confidence data Mean (SD)	Human annotation Mean (SD)	High-confidence data Mean (SD)
Silhouette	0.3702 (0.0101)	**0.3899** (0.0106)	0.3686 (0.0135)	**0.3927** (0.0149)
p-value	$p < 0.001$		$p < 0.001$	



(A-1) Human-annotation data of current economic conditions

(A-2) High-confidence data of current economic conditions

(B-1) Human-annotation data of future economic conditions

(B-2) High-confidence data of future economic conditions

Fig. 2. *t*-SNE visualization of inconsistencies in economic assessments and their justifications using raw data and extracted high-confidence data. Colors and shapes represent sentiment categories ranging from "greatly improved" to "greatly worse". More cohesive clusters in high-confidence data indicate improved label consistency.

published by the Cabinet Office of Japan. The *watcher* data reflects individuals' perceptions of economic conditions, while the *mer* data provides an official summary of ongoing economic trends and risk, serving as a critical resource for policymakers, businesses, and researchers. It also offers insights into government perspectives on fiscal and monetary policies, as well as the challenges facing Japan's economic landscape. Given that both datasets report on current conditions, it is posited that they should exhibit no significant differences in the sentiment. Collectively, the *watcher* and *mer* datasets comprise 493,460 and 47,911 sentences, respectively.

Table 5
Number of samples in training, validation and test data for each dataset type. "Current" and "Future" refer to datasets representing current and future conditions, respectively. "Current + Future" is a combined dataset including current conditions data and the extreme sentiment classes ("greatly improved" and "greatly worse") from the future conditions data.

	Training data (60%)	Validation data/Test data (20%)/(20%)	Total (100%)
Current	164,976	54,992	274,961
Future	178,083	59,361	296,806
Current + Future ("greatly improved", "greatly worse")	169,760	56,587	282,934

Table 6
Evaluation metrics for regression and classification performance of the proposed model on test data representing current economic conditions. The regression performance is reported as RMSE, while the classification performance is presented in terms of precision, recall, and F1-score for each sentiment category, along with the number of samples per class. Accuracy and macro-average scores are also provided.

Regression	RMSE	**0.1025**			
		Precision	Recall	F1-score	#Samples
Classification	Greatly Improved	**0**	**0**	**0**	120
	Slightly Improved	0.74	0.60	0.67	2,427
	No Change	0.86	0.86	0.86	26,807
	Slightly Worse	0.89	0.92	0.91	14,165
	Greatly Worse	0.73	0.73	0.73	11,473
	Accuracy	0.84			54,992
	Macro-average	**0.64**	**0.62**	**0.63**	

Table 7
Evaluation metrics for regression and classification performance of the proposed model on test data representing future economic conditions. The regression performance is reported as RMSE, while the classification performance is presented in terms of precision, recall, and F1-score for each sentiment category, along with the number of samples per class. Accuracy and macro-average scores are also provided.

Regression	RMSE	**0.0905**			
		Precision	Recall	F1-score	#Samples
Classification	Greatly Improved	**0.46**	**0.32**	**0.38**	135
	Slightly Improved	0.67	0.48	0.56	1,459
	No Change	0.90	0.91	0.91	32,734
	Slightly Worse	0.93	0.94	0.93	15,717
	Greatly Worse	0.75	0.73	0.74	9,317
	Accuracy	0.88			59,362
	Macro-average	**0.74**	**0.68**	**0.70**	

Table 8
Evaluation metrics for regression and classification performance of the proposed model on test data representing current economic conditions along with the "greatly improved" and "greatly worse" future economic conditions. The regression performance is reported as RMSE, while the classification performance is presented in terms of precision, recall, and F1-score for each sentiment category, along with the number of samples per class. Accuracy and macro-average scores are also provided.

Regression	RMSE	**0.1063**			
		Precision	Recall	F1-score	#Samples
Classification	Greatly Improved	**0.72**	**0.54**	**0.61**	257
	Slightly Improved	0.77	0.70	0.73	3,886
	No Change	0.86	0.85	0.85	26,806
	Slightly Worse	0.88	0.93	0.91	14,165
	Greatly Worse	0.71	0.72	0.72	11,473
	Accuracy	0.83			56,587
	Macro-average	**0.79**	**0.75**	**0.77**	

3.2. Baselines

To evaluate the performance of our proposed model, we selected two widely recognized sentiment analysis pre-trained language models as baselines: Jarvisx[1] and lxyuan.[2] This choice was based on a preliminary experiments that compared five sentiment analysis models for Japanese: Jarvisx, lxyuan, Christian,[3] Mizuiro,[4] and Kit.[5] Jarvisx and lxyuan were selected due to their closer alignment with human annotations and higher variance in sentiment scores.

- **Jarvisx**
  This state-of-the-art model is tailored for sentiment analysis within the economic domain. It boasts exceptional precision, evidenced by a Loss value of 0.0001 and perfect scores in both accuracy and F1-score (1.0). Jarvisx was fine-tuned on the chaABSA (Chakki Aspect-Based Sentiment Analysis) dataset, the first publicly available Japanese dataset for ABSA. This dataset includes 2,260 securities reports from 230 publicly listed companies during year 2016, comprising a total of 6,119 sentences. Within the dataset, 2,102 words were annotated as positive, negative, or neutral by professional annotators. Jarvisx outputs a sentiment category (positive or negative) with its possibility, ranging from 0 and 1. For negative sentiments, the possibilities were transformed using $1 - possibility$ to ensure the negative probabilities are lower than positive ones. Finally, the sentiment possibilities were normalized to a [0, 1] range, where 0.5 denotes neutrality.
- **lxyuan**
  Leveraging a DistilBERT-based architecture, this model excels in context-aware sentiment detection by incorporating attention mechanisms to highlight salient features in the input data. It was fine-tuned on a diverse, multilingual sentiment dataset containing 296,000 sentences from various sources, including Twitter, hotel reviews, movie critiques, and products reviews from Amazon. Lxyuan outputs possibilities for positive, neutral, and negative sentiments, with the highest possibility indicating the prevailing sentiment. Negative sentiments possibilities were transformed into negative values by multiplying by -1, and neutral possibilities were adjusted into zero. Finally, the sentiment scores were normalized to a range of $[-1, 1]$.

Using Jarvisx and lxyuan as baseline models provides a robust framework for economic sentiment analysis and superior contextual understanding. Evaluating our model against these baselines ensures a thorough assessment, highlighting its strengths in both general and context-specific sentiment analysis tasks.

3.3. Evaluation results

3.3.1. Generalization performance

For *watcher* and *mer* data, the sentiment score of each sentence was computed using Jarvisx, lxyuan, and the proposed model F-LUKE. Let $\mathbf{D}_{\text{watcher}}$ and $\mathbf{D}_{\text{mer}}$ represent the sentiment scores of the *watcher* and *mer* datasets, respectively. To quantify the similarity between $\mathbf{D}_{\text{watcher}}$ and $\mathbf{D}_{\text{mer}}$, we used the overlapping area of their distributions and the Jensen–Shannon (JS) divergence.

Firstly, the continuous probability density functions $W(x)$ for $\mathbf{D}_{\text{watcher}}$ and $M(x)$ for $\mathbf{D}_{\text{mer}}$ were computed using Eqs. (4)–(5).

$$W(x) = f_w(x), \quad M(x) = f_M(x) \tag{4}$$

$$f(x) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right) \tag{5}$$

where x_i is the original sentiment score, $K(\cdot)$ is the Gaussian kernel function, n is the number of sentences, and h is the bandwidth that controls the smoothness of the density function, selected using Scott's Rule. This process generates smooth, continuous density functions $W(x)$ and $M(x)$ over the data range.

The overlapping area of distributions. The overlapping region is determined by taking the smaller value of $W(x)$ and $M(x)$ at each point x_i. The density function $g(x)$ for the overlapping region is defined as Eq. (6).

$$g(x) = \min(W(x), M(x)) \tag{6}$$

The overlapping area is then calculated by integrating the overlapping density function $g(x)$ over the shared range as Eq. (7).

$$\begin{aligned} A &= \int_{x_{\min}}^{x_{\max}} g(x)\,dx = \int_{x_{\min}}^{x_{\max}} \min(W(x), M(x))\,dx \\ x_{\min} &= \min\left(\min(\mathbf{D}_{\text{watcher}}), \min(\mathbf{D}_{\text{mer}})\right) \\ x_{\max} &= \max\left(\max(\mathbf{D}_{\text{watcher}}), \max(\mathbf{D}_{\text{mer}})\right) \end{aligned} \tag{7}$$

Where $A = 1$ indicates complete overlap, while $A = 0$ signifies no overlap, providing a quantitative measure of the similarity between the distributions of $\mathbf{D}_{\text{watcher}}$ and $\mathbf{D}_{\text{mer}}$.

The JS divergence. To compute the JS divergence, the probability density functions must be evaluated on a common discretized grid to ensure $W(x)$ and $M(x)$ for numerical integration. Initially, a range of $[a, b]$ is defined to cover the values of $\mathbf{D}_{\text{watcher}}$ and $\mathbf{D}_{\text{mer}}$, and is divided into $N = 1000$ equally spaced points (Eqs. (8)–(9)).

$$x'_i = a + i\Delta x, \quad \Delta x = \frac{b-a}{N}, \quad i = 1, 2, \ldots, N \tag{8}$$

$$W(x) \approx f_W(x'_i), \quad M(x) \approx f_M(x'_i) \tag{9}$$

It is essential to ensure that $f_W(x'_i)$ and $f_M(x'_i)$ are valid probability distributions by normalizing their values so that their total sum equals 1 (Eqs. (10)–(11)).

$$W'(x'_i) = \frac{f_W(x'_i)}{\sum_{i=1}^{N} f_W(x'_i)\Delta x}, \quad M'(x'_i) = \frac{f_M(x'_i)}{\sum_{i=1}^{N} f_M(x'_i)\Delta x} \tag{10}$$

$$\sum_{i=1}^{N} W'(x'_i)\Delta x = 1, \quad \sum_{i=1}^{N} M'(x'_i)\Delta x = 1 \tag{11}$$

Subsequently, the mixture distribution $P(x'_i)$ is computed as the average of $W'(x')$ and $M'(x')$ (Eq. (12)).

$$P(x') = \frac{W'(x') + M'(x')}{2} \tag{12}$$

The Kullback–Leibler (KL) divergences between $W'(x')$ and $P(x')$, and between $M'(x')$ and $P(x')$, are calculated as Eqs. (13)–(14).

$$D_{\text{KL}}(W'||P) = \sum_{i=1}^{N} W'(x'_i)\log\frac{W'(x'_i)}{P(x'_i)}\Delta x \tag{13}$$

$$D_{\text{KL}}(M'||P) = \sum_{i=1}^{N} M'(x'_i)\log\frac{M'(x'_i)}{P(x'_i)}\Delta x \tag{14}$$

These two KL divergences are combined to compute the JS divergence as Eq. (15).

$$D_{\text{JS}}(W'||M') = \sqrt{\frac{D_{\text{KL}}(W'||P) + D_{\text{KL}}(M'||P)}{2}} \tag{15}$$

Where $D_{JS} = 1$ indicates that W and M are completely dissimilar (non-overlapping) distributions, while $D_{JS} = 0$ signifies identical distributions without divergence.

[1] https://huggingface.co/jarvisx17/japanese-sentiment-analysis
[2] https://huggingface.co/lxyuan/distilbert-base-multilingual-cased-sentiments-student
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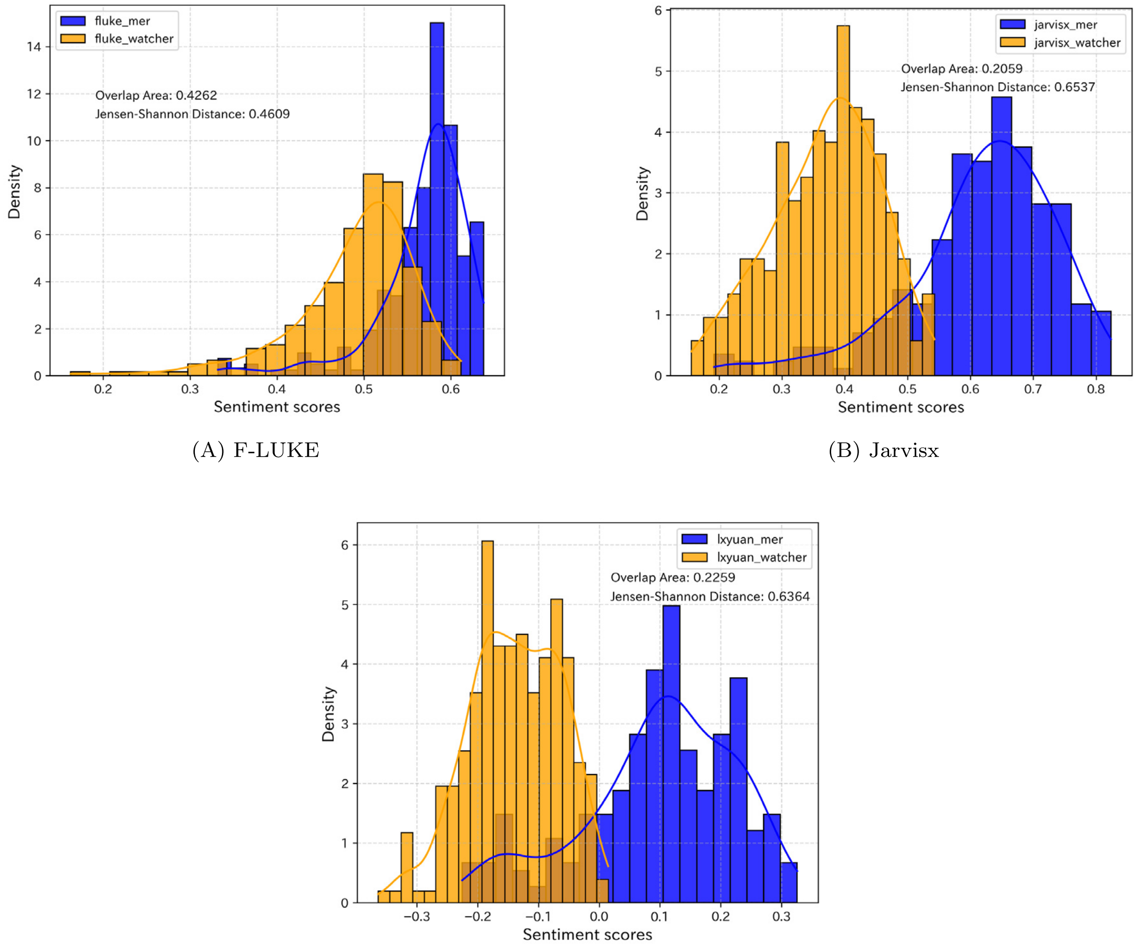
[4] https://huggingface.co/Mizuiro-sakura/luke-japanese-large-sentiment-analysis-wrime
[5] https://huggingface.co/kit-nlp/bert-base-japanese-sentiment-irony



(A) F-LUKE

(B) Jarvisx

(C) lxyuan

Fig. 3. Probability density distributions of sentiment scores calculated using F-LUKE, Jarvisx, and lxyuan. Each panel compares the score distributions for the *mer* and *watcher* datasets, with the overlap area and Jensen–Shannon distance reported to quantify distribution similarity.

Fig. 3 illustrates the probability density distributions of sentiment scores calculated using F-LUKE, Jarvisx, and lxyuan. F-LUKE exhibited an overlap area of 0.4262, in contrast to overlap areas of 0.2059 and 0.2259 for Jarvisx and lxyuan, respectively. Moreover, the JS divergence for F-LUKE was 0.4609, while Jarvisx and lxyuan recorded 0.6537 and 0.6364, respectively. These results indicated that F-LUKE demonstrated greater robustness across different datasets, maintaining consistent performance even when with varied data inputs, which underscores its superior generalization capability.

3.3.2. *Consistence with other metrics*

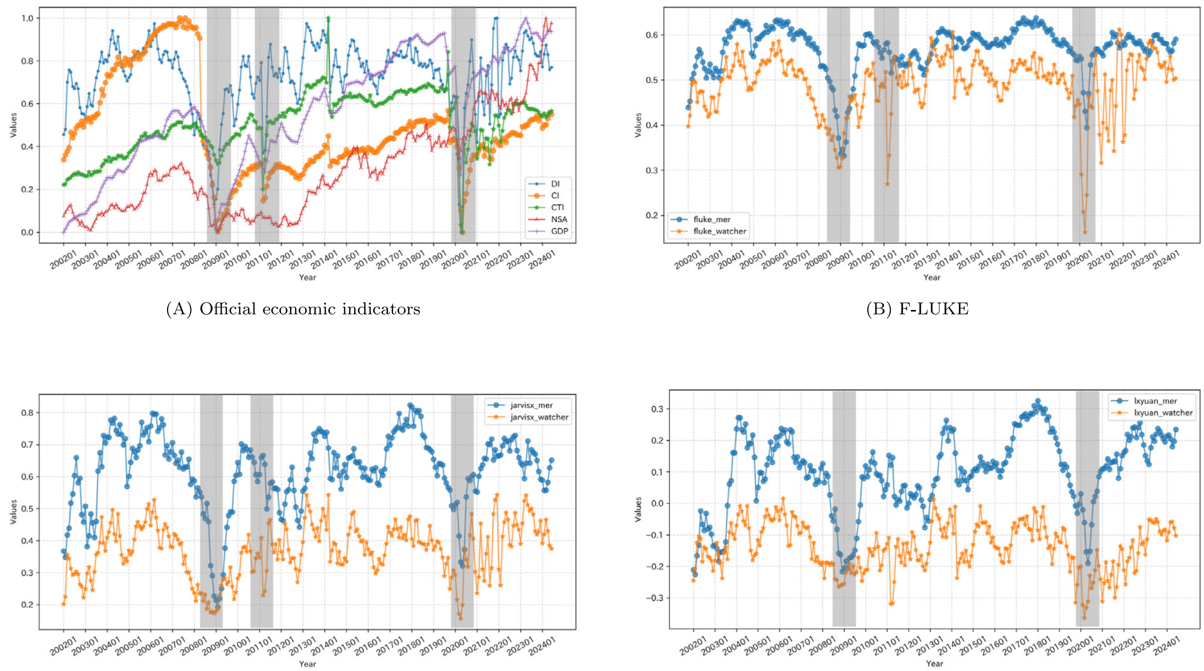
Various official economic indicators are designed to reflect the economic conditions, and synchronized movements among them are considered desirable. This section assesses the consistency between sentiment scores calculated by F-LUKE, Jarvisx, and lxyuan, and five official economic indicators: the DI, the composite index (CI—an overall economic indicator derived by taking the weighted average of multiple economic indicators), the consumption trend index (CTI—representing the average monthly total consumption expenditure of household), the Nikkei stock average (NSA—a leading indicator of economic conditions as stock prices typically decline when the economy worsens), and the GDP (regarded as the most comprehensive economic indicator). Note that because GDP are reported on a quarterly basis, linear interpolation was performed to obtain corresponding monthly values, allowing alignment with the other indicators. Specifically, the average sentiment score for all sentences in each month was computed to generate the sentiment index for that month, which was visualized alongside the temporal variations of official indicators in Fig. 4.

The official economic indicators highlighted three significant economic recessions in Japan since 2002. The first recession occurred in late 2008 when the collapse of Lehman Brothers triggered a global financial crisis that severely impacted Japan's economy. The second recession followed the Great East Japan Earthquake in early 2011, one of the most devastating natural disasters in Japan's history. The third recession began in early 2020 with the onset of the COVID-19 pandemic, which had far-reaching impacts on Japan. Consistently, F-LUKE and Jarvisx accurately reflected these economic downturns, whereas lxyuan, based on the *mer* data, indicated an increasing trend in early 2011.

Moreover, the Euclidean distance and maximum information coefficient (MIC) were used to quantify the similarity between the sentiment indices derived from the *mer* and *watcher* data. For F-LUKE, Jarvisx, and lxyuan, the Euclidean distances were 1.56, 4.33, and 4.24, respectively, while the MIC values were 0.4929, 0.4630, and 0.4092, respectively. These results suggest that the sentiment index of F-LUKE not only reflected the fluctuations of other economic indicators but was also less influenced by data diversity.

4. Relationship between quantitative and qualitative economic indicators

Economic indicators can be broadly categorized into two types: quantitative and qualitative. Quantitative indicators are measurable



(A) Official economic indicators (B) F-LUKE

(C) Jarvisx (D) lxyuan

Fig. 4. Temporal variations in official economic indicators and sentiment indices derived from F-LUKE, Jarvisx, and lxyuan. Shaded areas represent periods of economic recession.

and reflect objective economic conditions, whereas qualitative indicators capture subjective perceptions of economic conditions. Despite their distinct nature and methodology, the interplay between these two types of indicators can yield a deeper understanding of economic trends. This section examines how qualitative indicators (DI, sentiment indices calculated by F-LUKE, Jarvisx, and lxyuan) can complement quantitative indicators (CI, CTI, NSA) in capturing shifts in economic conditions as indicated by GDP.

Initially, exploratory factor analysis (EFA) was employed to determine the number of latent factors within the data and to evaluate whether qualitative and quantitative indicators represent distinct latent factors. Selection of factors was based on eigenvalues exceeding 1. The analysis revealed three latent factors, with the primary five eigenvalues being 6.4956, 1.8195, 1.1539, 0.6514, and 0.4435. Subsequently, EFA with three latent factors confirmed the results, which are detailed in Table 9.

Factor 1 (F1) includes Jarvisx_watcher, lxyuan_watcher, F-LUKE_watcher, and DI; Factor 2 (F2) includes Jarvisx_mer, lxyuan_mer, and F-LUKE_mer; Factor 3 (F3) includes CTI, NSA, and GDP. These results demonstrated that qualitative and quantitative indicators were categorized into distinct factors, with qualitative indicators further segmented by their data source. Notably, the communality of CI was relatively low at 0.2828, and its highest factor loading was 0.4853 for F2, significantly lower than those of other indicators (communality: 0.6258–0.9950; the highest three factor loadings for F2: 0.7475–0.8887). This suggests that CI shares limited commonality with the other observed variables.

Based on the identification of these three latent factors and their associated indicators, a model was developed using structural equation modeling (SEM) to explore the interactions between qualitative and quantitative indicators and their respective impacts on GDP. Due to its low communality and factor loading, CI was excluded from the analysis. The results of SEM for both the qualitative model and the combined model (quantitative + qualitative) are presented in Table 10, with the corresponding path diagrams illustrated in Fig. 5. The model fit indices are detailed in Table 11.

Table 9
Results of exploratory factor analysis (EFA), showing communalities and factor loadings for three extracted factors (F1, F2, F3). The "Communality" column indicates the proportion of variance in each variable explained by the factor model. Columns F1, F2, and F3 represent the factor loadings of each variable on the respective factors. The highest and lowest factor loadings for each economic indicator are highlighted in bold and in italics for clarity. $N = 270$, representing monthly economic metrics from January 2002 to June 2024. Abbreviations: DI = Diffusion Index; CI = Composite Index; CTI = Consumption trend Index; NSA = Nikkei Stock Average; GDP = Gross Domestic Product.

	Communality	F1	F2	F3
Jarvisx_mer	0.9854	0.3557	**0.8887**	0.2629
lxyuan_mer	0.8908	0.2983	**0.7475**	0.4930
F-LUKE_mer	0.9187	0.3850	**0.8315**	0.2812
Jarvisx_watcher	0.8798	**0.8300**	0.3301	0.2862
lxyuan_watcher	0.7391	**0.7937**	0.3245	0.0621
F-LUKE_watcher	0.9950	**0.9403**	0.2925	0.1589
DI	0.9683	**0.8787**	0.3963	0.1979
CI	*0.2828*	*0.2058*	*0.4853*	*−0.0697*
CTI	0.6258	0.3727	0.1202	**0.6873**
NSA	0.6951	0.0296	0.0873	**0.8286**
GDP	0.9951	0.1325	0.1953	**0.9692**

According to Table 10, the combined model illustrated that GDP was significantly impacted by both quantitative and qualitative indicators. However, the coefficient for qualitative_watcher ($\beta = -0.1516$) showed a negative relationship with GDP. In the qualitative-only model, the coefficient β for qualitative_mer was significantly positive at 0.4937 ($p = 0$), indicating a strong and significant impact. In contrast, qualitative_watcher exhibited a much smaller coefficient ($\beta = 0.0254$) and did not reach statistical significance ($p = 0.7381$). These results

Table 10
SEM results: relationships between latent variables and observed variables in qualitative-only and combined (quantitative + qualitative) models. β is the path coefficient, and p indicates the statistical significance ($p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), $p < 0.0001$ (****)). $N = 270$, representing monthly economic metrics from January 2002 to June 2024.

Latent variables	Observed variables	Qualitative		Quantitative + Qualitative	
		β	p	β	p
GDP	Quantitative	–	–	0.9908	****
	Qualitative_mer	0.4937	****	0.1414	**
	Qualitative_watcher	0.0254	0.7381	−0.1516	**
Qualitative	Qualitative_mer	0.8691	–	0.8898	–
	Qualitative_watcher	0.8139	1	0.7952	****
Quantitative	NSA	–	–	0.8212	–
	CTI	–	–	0.7611	****
Qualitative_m	lxyuan_mer	0.9131	****	0.9129	****
	Fluke_mer	0.9619	****	0.9619	****
	Jarvisx_mer	0.9857	–	0.9858	–
Qualitative_w	DI	0.9874	–	0.9876	–
	lxyuan_watcher	0.8468	****	0.8465	****
	Fluke_watcher	0.9852	****	0.9851	****
	Jarvisx_watcher	0.9381	****	0.9381	****

Table 11
Model fit indices for qualitative-only and combined (quantitative + qualitative) SEM models. CFI is the Comparative Fit Index, GFI is the Goodness of Fit Index, AGFI is the Adjusted Goodness of Fit Index, NFI is the Normed Fit Index, TLI is the Tucker-Lewis Index, RMSEA is the Root Mean Square Error of Approximation, AIC is the Akaike Information Criterion, and BIC is the Bayesian Information Criterion. Bold values indicate the model with superior performance for each metric.

Model fit indices	Qualitative	Quantitative + Qualitative
CFI	**0.8836**	0.8399
GFI	**0.8796**	0.8346
AGFI	**0.8017**	0.7519
NFI	**0.8796**	0.8346
TLI	**0.8083**	0.7599
RMSEA	0.2937	**0.2886**
AIC	**34.9525**	44.7987
BIC	**103.3225**	134.7592

suggested that official statements, as captured in *mer* data, contribute more significantly to explaining variations in GDP.

The objective of SEM is to replicate the covariance matrix of the data as accurately as possible while minimizing the number of parameters used in the model. The adequacy of model fit is gauged by various metrics, and it is crucial to assess from multiple perspectives to achieve a comprehensive evaluation. In this study, 8 model fit indices were employed, with detailed explanations provided in Table 12.

The evaluation results for the model using only qualitative indicators and the model combining both qualitative and quantitative indicators are summarized in Table 11. Although the combined model showed a slight advantage in RMSEA, the qualitative-only model generally exhibited better fit indices across the remaining 7 measures. Overall, the qualitative-only model demonstrated superior fit.

Fig. 5 illustrates why the inclusion of quantitative indicators deteriorated model fit. As shown, there was a strong positive correlation between qualitative and quantitative indicators ($\beta = 0.5413, p = 0$). Moreover, quantitative indicators had a significant direct impact on GDP ($\beta = 0.9908, p = 0$), indicating that the inclusion of quantitative indicators caused some effects of qualitative indicators to influence GDP indirectly through quantitative pathways. This diagram underscored the transformation of qualitative assessments into measurable quantitative indicators, reflecting their crucial role in economic forecasting and emphasizing the significance of qualitative factors in predicting economic conditions.

5. Conclusions

The economy is fundamentally driven by the daily activities and sentiments of the public, who are its primary agents. These sentiments inherently influence consumption behaviors, thereby serving as a driving force for economic activity. Thus, a thorough understanding of the psychology and sentiments underlying people's daily lives is essential for accurately evaluating current economic conditions and predicting future economic trends. Given this context, economic sentiment analysis is gaining increasing prominence.

The development of sentiment analysis models typically involves the utilization of benchmark datasets (e.g. the Economy Watchers Survey data for Japanese, the Financial PhraseBank dataset for English), which are widely recognized and commonly employed. However, concerns regarding the accuracy of these datasets continue to arise, primarily due to psychological biases and inconsistencies in human annotations.

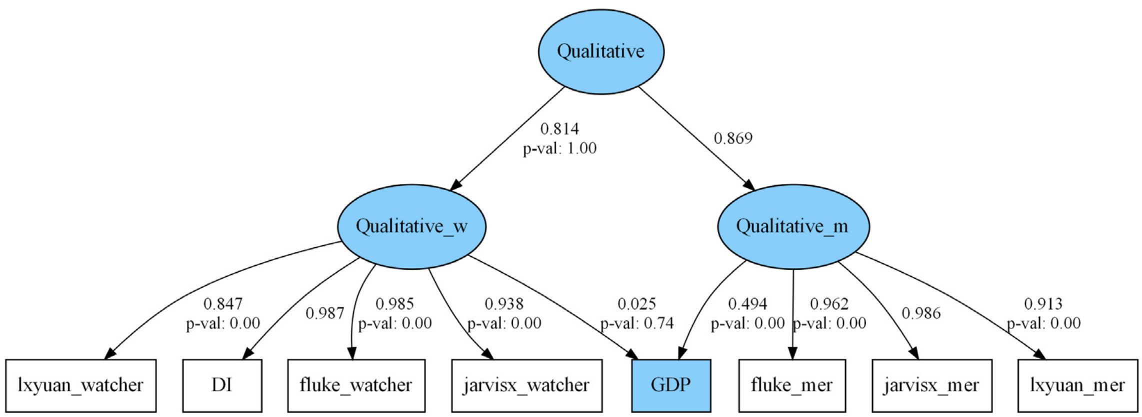
To address this issue, this study introduced a data filtering approach designed to enhance the reliability of data. Subsequently, a robust sentiment analysis model tailored to the Japanese economy was developed. The results demonstrated that the proposed model exhibited superior generalization capability across various datasets and effectively captured fluctuations in other official quantitative economic indicators. Moreover, the findings revealed a statistically significant correlation between qualitative and quantitative economic indicators, highlighting the potential of qualitative factors in predicting economic conditions. Given the low cost and widespread availability of textual data, along with advancements in NLP techniques, the integration of qualitative data into economic analysis holds considerable promise. This approach could not only improve the accuracy and robustness of economic predictions but also provide a more comprehensive understanding of economic dynamics.

However, this study has certain limitations. The evaluation was conducted using the Economy Watchers Survey, which is highly reliable and representative of the Japanese economy, the absence of other comparable benchmark datasets in Japanese limits our ability to validate the generalizability of the proposed approach across different data sources.

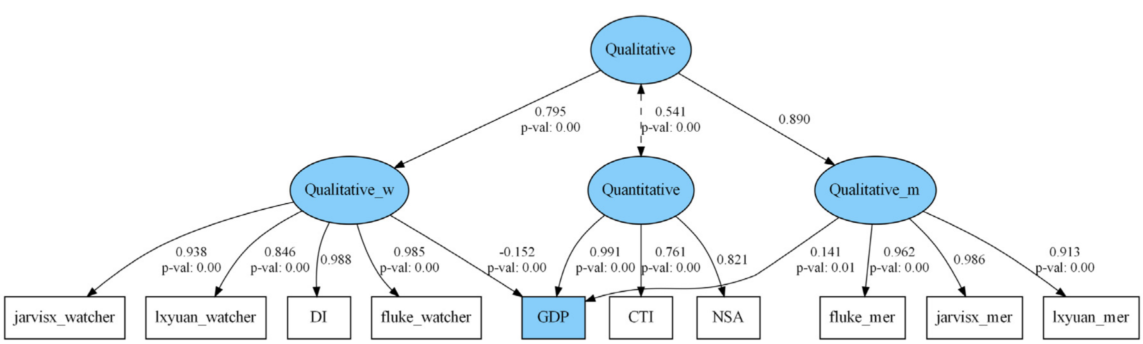
Future research should address this limitation by constructing or identifying additional datasets to enable broader validation analyses. Moreover, besides achieving high accuracy of economic sentiment analysis models, it is essential to deepen the understanding of the underlying mechanisms that influence economic dynamics.

Table 12
Model fit indices for evaluating model-data fit. The table lists each index, its abbreviation, and a brief explanation of its purpose.

Model fit index	Abbreviation	Explanation
Comparative Fit Index	CFI	Compare the fit of the proposed model against a null model, which assumes that all variables are uncorrelated.
Goodness of Fit Index Adjusted Goodness of Fit Index	GFI AGFI	Measure the proportion of variance explained by the model.
Normed Fit Index Tucker-Lewis Index	NFI TLI	Compare the fit of the proposed model against a baseline model.
Root Mean Square Error of Approximation	RMSEA	Assesses the degree of approximation error per degree of freedom in the proposed model.
Akaike Information Criterion Bayesian Information Criterion	AIC BIC	Information criteria based.



(A) Qualitative



(B) Quantitative + Qualitative

Fig. 5. Structural Equation Modeling (SEM) path diagrams for (A) the qualitative model and (B) the combined quantitative + qualitative model. The diagrams display standardized path coefficients with associated *p*-values, illustrating the relationships between latent variables (blue ovals) and observed variables (rectangles).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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